MATHEMATICAL MODELING DOCUMENT

From Unit Economics to Free Cash Flow

A Mathematical Model of Financing-Driven Expansion in AI Data Centers

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VERSION 1.0 | 11 NOVEMBER 2026

Structured edition based on the supplied 8-page manuscript

Purpose: formalize the operating-cash-flow, leverage, collateral, timing, and stochastic constraints that govern expansion financing for AI data centers.

Executive Summary

This document organizes the supplied model into a reproducible analytical framework. The baseline model compares unit EBITDA with unit capital expenditure and debt service; the extended model then adds utilization dependence, residual-value risk, endogenous credit spreads, construction delays, revenue-mix volatility, off-balance-sheet structures, and stochastic uncertainty.

> **Central result**
>
> Under the stated baseline inputs (unit EBITDA of 500, unit capex of 3,950, debt of 2,500 per MW, and an 8.5% debt cost), the critical annual expansion rate is approximately 7.3% of the installed base. The unlevered threshold is approximately 12.7%. The manuscript compares these thresholds with an industry expansion range of 50%-90%.

Main model outputs

Output	Baseline / reported result	Interpretation
Critical expansion speed	7.3% levered; 12.7% unlevered	Operating EBITDA less interest cannot fund capex above this rate under the stated identity.
Unit EBIT	-59.6 ($10k/MW/year)	Depreciation exceeds EBITDA, so unit economics are loss-making on a GAAP EBIT basis.
Facility FCF	421 ($10k/MW/year)	EBITDA less maintenance capex at 2% of unit capex.
DSCR at full utilization	0.58 / 0.65 / 0.81 across the three tested structures	All are below the 1.20 covenant threshold.
Low-state collateral coverage	0.08 without guarantee; 0.18 with guarantee	Both are materially below the 1.20 refinancing threshold.
Quarterly DSCR below 1.0	26%-32% (normal approximation)	Reported model output under the training/inference volatility assumptions.

Document map

- Part I - Symbols, time structure, and modeling boundaries
- Part II - Baseline unit economics, company FCF, and expansion threshold
- Part III - Nonlinear operating, covenant, collateral, and timing constraints
- Part IV - Stochastic specification and sensitivity results
- Part V - Calibration comparisons, limitations, and consolidated conclusion
- Appendix - Reproduction formulas and interpretation notes

Source discipline: numerical inputs and claims are retained from the supplied manuscript. Additional arithmetic checks and interpretation notes are explicitly identified; no external market data have been added.

I. Model Scope, Notation, and Timeline

1.1 Model objective and architecture

The model asks whether an AI data-center operator can finance rapid capacity growth from operating cash flow, or whether debt, asset-backed structures, guarantees, and refinancing are required to bridge the gap between EBITDA generation and capital expenditure. Its analytical architecture is:

Layer	Mechanism	Primary output
1. Baseline identity	Unit economics, financing cost, annual expansion capex	Critical expansion speed μ*
2. Operating state	Utilization-dependent EBITDA and fixed O&M	DSCR(u)
3. Asset / refinancing	Residual-value step risk and guarantees	Collateral coverage at maturity
4. Timing	Construction ramp and loan maturity	Interim cash leakage / refinancing cliff
5. Revenue volatility	Training/inference mix and quarterly variation	Probability of DSCR falling below 1.0
6. Uncertainty	Joint random variables and sensitivity axes	Default / infeasibility probabilities

1.2 Symbol table

Category	Symbol	Definition	Unit
Scale	n	Installed equivalent capacity	MW
Scale	m	Annual new capacity	MW/year
Capital	C0	Unit capital expenditure	10k USD/MW
Capital	D	Annual depreciation	10k USD/MW/year
Capital	s	Residual-value rate	ratio
Capital	Tdep	Depreciation life	years
Operations	R1	Annual full-load unit revenue	10k USD/MW/year
Operations	E	Annual electricity cost	10k USD/MW/year
Operations	α	Variable O&M share of revenue	ratio
Operations	β	Fixed O&M share of full-load revenue	ratio
Operations	γ	Additional / unlisted cost share	ratio
Operations	u	Utilization	ratio
Operations	PUE	Power usage effectiveness	ratio
Operations	p_elec	Industrial electricity price	USD/kWh
Financing	B	Outstanding debt	10k USD
Financing	b = B/n	Unit debt	10k USD/MW
Financing	r_d	Cost of debt	annual rate
Financing	r_f	Risk-free rate	annual rate

Category	Symbol	Definition	Unit
Financing	σ	Credit spread	basis points
Financing	λ	On-balance-sheet debt share	ratio
Covenants	DSCR*	Reference / covenant DSCR	ratio
Covenants	θ	Minimum collateral coverage	ratio
Timing	Tloan	Loan maturity / amortization horizon	years
Timing	τ	Construction period	months
Revenue mix	θ_inf	Inference revenue share	ratio
Stochastic	s~	Random residual-value rate	ratio
Stochastic	p_tok	Token unit price	USD / million tokens
Stochastic	p_GPU	GPU rental rate	USD/hour
Policy	τ_tax	Effective tax rate	ratio
Policy	g	Residual-value guarantee ratio	ratio

Notation note

The source manuscript contains duplicated or corrupted symbol strings in several equations and tables. This edition normalizes them to the symbols listed above. The source does not fully define P_IT in the electricity-cost expression, nor does it provide all unit-conversion details.

1.3 Timeline

Stage	Interval	Cash-flow / model treatment
Construction	t ∈ [-τ, 0]	Capital expenditure occurs; depreciation and interest begin from equipment delivery; revenue is zero.
Ramp	t ∈ [0, τ_r]	Revenue increases linearly or under an S-curve toward steady state.
Steady state	t ∈ [τ_r, Tloan]	EBITDA is generated according to utilization u.
Refinancing / exit	t = Tloan	Principal matures; collateral is valued at the realized residual-value rate; maturity coverage is tested.

1.4 Modeling boundaries

The baseline focuses on operating and financing cash flows. It does not fully model intertemporal tax optimization, equity dilution, working-capital fluctuations, regional electricity-price and PUE heterogeneity, or GPU-generation mix. Later sections identify further limitations in revenue endogeneity, agent heterogeneity, aggregation frequency, and strategic responses.

II. Baseline Model: Unit Economics and Expansion Threshold

2.1 Annualized cost and EBITDA

ANNUAL DEPRECIATION $D = C0 \times (1 - s) / Tdep$

ANNUAL ELECTRICITY COST RELATIONSHIP $E = P_IT \times 8{,}760 \times PUE \times p_elec$

The manuscript presents this relationship but does not define P_IT or show the complete conversion into 10k USD/MW/year.

ADJUSTED FULL-LOAD EBITDA $EBITDA_1(full\ load) = R1 \times (1 - \alpha - \gamma) - E$

The manuscript first calculates EBITDA before making the additional cost load explicit, then introduces γ to reconcile the result to a 50% EBITDA margin.

Input / calculation	Value	Unit / explanation
C0	3,950	10k USD/MW
s	0.15	Residual value rate
Tdep	6	years
D = 3,950 × 0.85 / 6	559.6	10k USD/MW/year
R1	1,000	10k USD/MW/year
α	0.12	Variable O&M share
E	80.6	10k USD/MW/year
Unadjusted EBITDA = 1,000 × 0.88 - 80.6	799.4	10k USD/MW/year
Implied additional cost γR1	299.4	10k USD/MW/year
γ (rounded)	0.30	Additional cost share
Adopted EBITDA_1	500	10k USD/MW/year; 50% margin

Baseline convention

After estimating γ ≈ 0.30, the manuscript fixes EBITDA_1 = 500 as the baseline input. Subsequent baseline calculations use 500 rather than the unadjusted 799.4.

2.2 Unit EBIT and facility FCF

UNIT EBIT $EBIT_1 = EBITDA_1 - D = 500 - 559.6 = -59.6$

STEADY-STATE FACILITY FCF $FCF_facility = EBITDA_1 - Maintenance\ Capex = 500 - 0.02 \times 3{,}950 = 421$

The unit is loss-making on a GAAP EBIT basis because annual depreciation exceeds EBITDA. Under the manuscript assumption that maintenance capex equals 2% of C0, steady-state facility FCF is 421 (10k USD/MW/year).

2.3 Company-level FCF identity

COMPANY-LEVEL ANNUAL FCF IDENTITY $$FCF_{company} = n \times EBITDA_1 - m \times C0 - r_d \times b \times n$$

Here n is installed capacity, m is annual new capacity, C0 is unit capex, and $r_d \times b \times n$ is annual interest expense on the modeled debt balance. The identity is the source model's baseline company-level FCF definition.

FCF definition boundary

The company-level identity above does not deduct the maintenance capex used in FCF_facility, and it does not explicitly deduct taxes or working-capital investment. The two FCF measures therefore have different scopes in the source model and should not be treated as interchangeable.

2.4 Critical expansion speed

CRITICAL EXPANSION SPEED $$\mu^* = (m/n)^* = (EBITDA_1 - r_d \times b) / C0$$

Financing case	Unit interest $r_d \times b$	Critical expansion μ*
Unlevered (b = 0)	0	500 / 3,950 = 12.7%
Levered (b = 2,500; r_d = 8.5%)	212.5	(500 - 212.5) / 3,950 = 7.3%

The manuscript compares the 7%-13% critical range with industry expansion of approximately 50%-90%. Under the model identity, growth materially above μ* requires funding beyond the modeled operating cash flow.

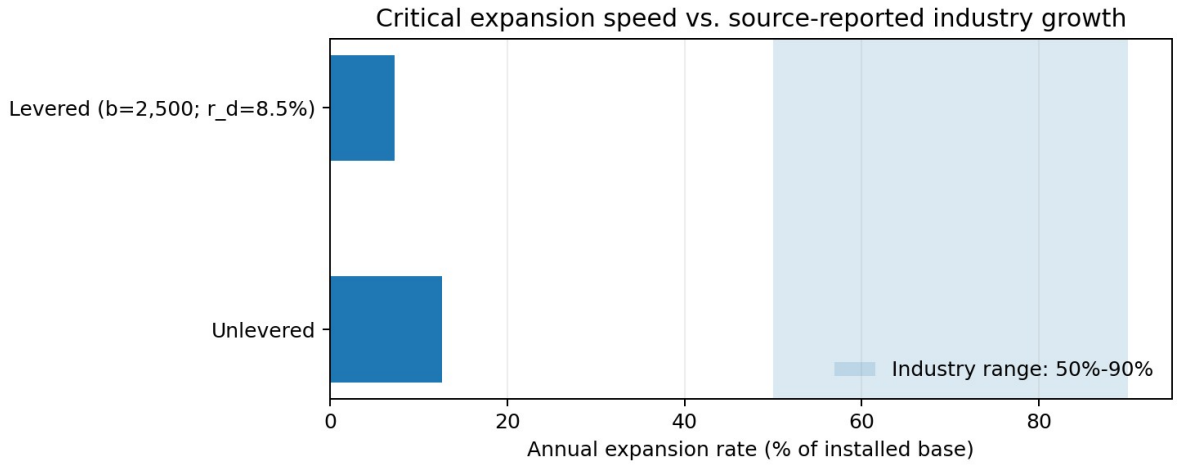


Figure 1. Expansion thresholds calculated from the manuscript inputs; the 50%-90% range is the manuscript's industry comparison.

Interpretation

This is a static annual identity, not a full multi-period financing optimization. It isolates the gap between cash generated by the installed base and capex required for new capacity.

III. Extended Model: Operating and Financing Constraints

3.1 Utilization-dependent EBITDA and DSCR

The extension separates fixed O&M from utilization-sensitive operating costs. With β = 0.05, γ = 0.30, α = 0.12, R1 = 1,000, and E = 80.6:

UTILIZATION-DEPENDENT EBITDA $$EBITDA_1(u) = R1 \times (1 - \alpha - \gamma) \times u - E \times u - \beta \times R1$$

SIMPLIFIED EXPRESSION $$EBITDA_1(u) = (1{,}000 \times 0.58)u - 80.6u - 50 = 499.4u - 50$$

DEBT-SERVICE COVERAGE $$DSCR(u) = (499.4u - 50) / (r_d \times b + A), \quad A = C0 \times (1 - s) / Tloan$$

The source uses A as annual principal amortization. For the numerical cases below, A is computed using the stated loan horizon and residual value assumptions.

Debt structure	Interest r_d×b	Principal A	Debt service denominator	DSCR at u=1	u* for DSCR=1.20
b=2,500; Tloan=6	212.5	559.6	772.1	0.58	1.96
b=1,580; Tloan=6	134.3	559.6	693.9	0.65	1.77
b=1,580; Tloan=8	134.3	419.7	554.0	0.81	1.43

All three computed utilization thresholds exceed 1.0, meaning the DSCR covenant of 1.20 cannot be met even at full utilization under the tested assumptions. The manuscript interprets this as evidence that the baseline debt structure is not sustainable without materially lower leverage, longer amortization, or institutional credit support.

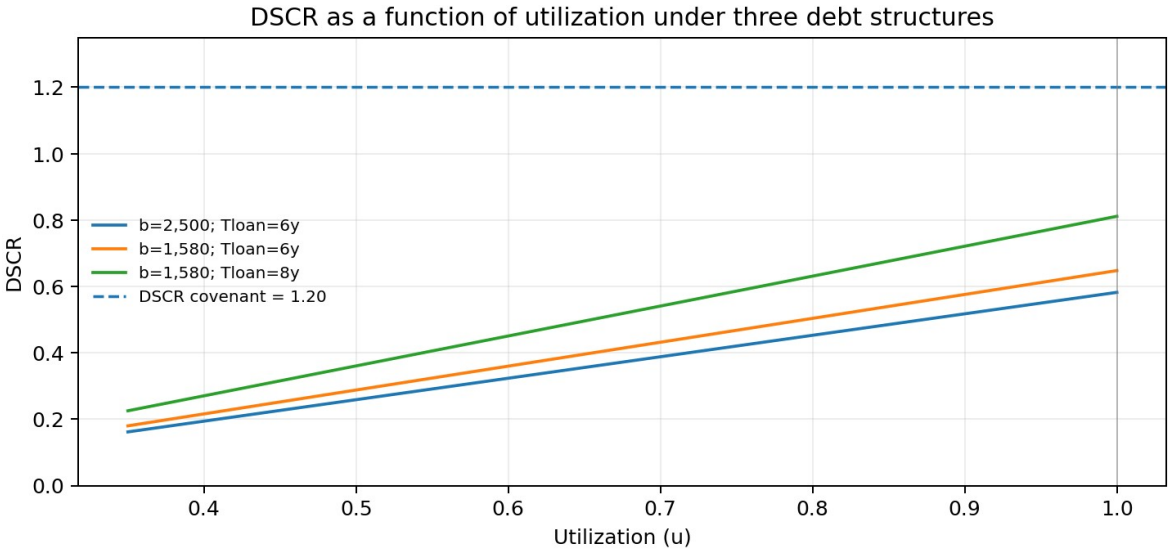


Figure 2. DSCR curves derived from EBITDA_1(u) = 499.4u - 50 and the three stated debt-service denominators.

3.2 Residual-value step risk and guarantees

The residual-value rate is modeled as a two-state random variable: $s \sim \in \{0.40, 0.05\}$, with probability 0.60 on the high state. The expected residual-value rate is therefore 0.26.

EXPECTED RESIDUAL VALUE $$E[s\sim] = 0.60 \times 0.40 + 0.40 \times 0.05 = 0.26$$

GUARANTEED EFFECTIVE COLLATERAL VALUE $$V_{eff}(s\sim) = C0 \times s\sim + g \times C0 \times \max(0, s_{ref} - s\sim)$$

With C0 = 3,950, guarantee ratio g = 0.25, reference residual value s_ref = 0.30, and low-state residual value s_L = 0.05:

Low-state collateral calculation	Value
Unprotected asset value: 3,950 × 0.05	197.5
Guarantee contribution: 0.25 × 3,950 × (0.30 - 0.05)	246.9
Effective collateral V_eff	444.4
Coverage without guarantee: 197.5 / 2,500	0.08
Coverage with guarantee: 444.4 / 2,500	0.18
Required coverage θ	1.20

The guarantee more than doubles low-state collateral coverage, but the resulting 0.18 coverage remains far below the 1.20 threshold. The manuscript therefore treats guarantees as a marginal improvement rather than a structural solution.

3.3 Endogenous financing costs and refinancing cliff

CREDIT SPREAD FEEDBACK $$\sigma_{bp}(DSCR) = \sigma 0 \times (DSCR_{ref} / DSCR)^{\eta}, \quad r_d = r_f + \sigma_{bp} / 10{,}000$$

The manuscript specifies σ0 = 200 basis points, DSCR_ref = 1.5, and η = 2. Because σ is denominated in basis points in the source symbol table, this edition writes the conversion explicitly when adding the spread to a decimal interest rate. The functional form makes the credit spread rise as DSCR deteriorates, feeding weaker coverage back into a higher cost of debt.

MATURITY COLLATERAL COVERAGE CONSTRAINT $$V_{eff}(s\sim(T_{loan})) / B_{remaining}(T_{loan}) \geq \theta$$

If principal amortizes evenly, B_remaining at maturity can approach zero and the collateral test becomes a repayment test. With a balloon structure, a substantial balance remains at maturity and low residual value can block refinancing. The source identifies this maturity-node collateral test as a potential trigger before a conventional annual cash-flow default.

DUAL-CONSTRAINT FEASIBLE REGION $$F = \{(u,b,T_{loan},s\sim): DSCR(u,b,T_{loan}) \geq 1.20 \text{ and } V_{eff}(s\sim)/B_{remaining}(T_{loan}) \geq \theta\}$$

Refinancing cliff

The model's feasible region is the intersection of an operating covenant and a collateral coverage condition. A project may satisfy one constraint and still fail the other; the manuscript highlights low-state collateral coverage as a binding refinancing constraint.

3.4 Construction ramp delay

During a construction period of τ months, depreciation and interest accrue while revenue is zero. The manuscript defines first-year leakage as:

CONSTRUCTION-PERIOD CASH LEAKAGE $$L(\tau) = (\tau/12) \times (D + r_d \times b) \times n + (\tau/12) \times C0 \times m$$

The first term represents depreciation and interest accrual on installed capacity during the delay. The second term scales with new capacity under construction. The manuscript reports that at m/n = 0.5 and τ = 4 months, leakage is approximately 51% of annual EBITDA.

Arithmetic / scope reconciliation

Using only the first term gives (4/12) × (559.6 + 212.5) / 500 ≈ 51.5%, matching the stated 51%. Including the second term in the displayed full formula at m/n = 0.5 adds (4/12) × 3,950 × 0.5 / 500 ≈ 131.7 percentage points. Thus, the 51% statement appears to quantify the installed-base depreciation/interest component only; the manuscript does not explain whether the new-capacity term is excluded from that percentage comparison.

3.5 Training / inference maturity mismatch

Quarterly EBITDA variability is modeled as a weighted combination of inference and training revenue volatility:

QUARTERLY COEFFICIENT OF VARIATION $$\sigma_Q = \sqrt{[\ \theta_{inf}^2\sigma_{inf}^2 + (1-\theta_{inf})^2\sigma_{train}^2 + 2\theta_{inf}(1-\theta_{inf})\rho\sigma_{inf}\sigma_{train}\ ]}$$

For θ_inf = 0.60, σ_inf = 0.10, σ_train = 0.60, and correlation ρ = 0.20, the model obtains σ_Q ≈ 0.26. At θ_inf = 0.40, σ_Q ≈ 0.37.

NORMAL APPROXIMATION $$P(DSCR_Q < 1.0) = \Phi[(1.0 - mean(DSCR_Q)) / (mean(DSCR_Q) \times \sigma_Q)]$$

At mean quarterly DSCR of 1.20, the manuscript reports a probability of approximately 26% when σ_Q = 0.26, rising to approximately 32% when σ_Q = 0.37. These are approximation outputs, not empirical default frequencies.

3.6 Off-balance-sheet structure and tax timing

The on-balance-sheet share λ allocates λB to the parent and (1-λ)B to SPVs. Parent credit quality affects SPV rent or financing pressure, while SPV DSCR deterioration can feed back into the parent's implicit obligation valuation.

COUPLED PARENT / SPV FINANCING RELATIONSHIPS $$r_{d,parent} = f(\lambda B / EBITDA_{parent}) \quad ; \quad r_{d,SPV} = g(DSCR_{SPV}, \lambda)$$

ACCELERATED-DEPRECIATION TAX SHIELD $$T_{shield} = \tau_{tax} \times D_{accelerated}$$

The source states that first-year tax shield can reach τ_tax × C0 ≈ 830 (10k USD/MW), or about 1.66 times EBITDA. The manuscript treats this as a timing shift in FCF rather than a change in lifetime tax burden or the long-run EBITDA-to-capex ratio.

IV. Stochastic Treatment and Sensitivity

4.1 Stochastic model specification

The source identifies four key jointly distributed variables: utilization u, residual value s~, token price p_tok, and cost of debt r_d.

> JOINT STOCHASTIC STATE **X = (u, s~, p_tok, r_d)**

Variable	Distribution / process	Parameters supplied
u	Beta distribution	Mean 0.85; standard deviation 0.12
s~	Two-state discrete distribution	s_H = 0.40; s_L = 0.05; P(s_H)=0.60
p_tok	Geometric Brownian motion	Drift -15% per year; volatility 25%
r_d	Endogenous financing rate	Determined by DSCR with random disturbances

> **Optional moment-matched Beta parameters**
>
> If a standard Beta distribution is intended and the supplied mean / standard deviation are matched exactly, the implied parameters are approximately $\alpha_u = 6.68$ and $\beta_u = 1.18$. These are derived values, not parameters explicitly printed in the source manuscript.

4.2 Monte Carlo output metrics

- P(FCF_company < 0): probability that company-level FCF is negative.
- P(DSCR < 1.0): probability of operating covenant distress.
- P(V_eff / B_remaining < θ): probability of a maturity collateral shortfall.
- P(DSCR < 1 or V_eff / B_remaining < θ): joint default / infeasibility probability.

The manuscript argues that joint default risk can materially exceed single-variable point estimates because utilization decline, residual-value step-down, and credit-spread widening are positively correlated.

> **Reproducibility boundary**
>
> The supplied manuscript specifies the stochastic variables and output metrics, but does not provide a covariance matrix, shock equations, simulation horizon, number of Monte Carlo draws, random seed, or numerical simulation results. Consequently, this edition documents the stochastic design but does not claim to reproduce a completed Monte Carlo run.

4.3 Sensitivity results reported in the source

Sensitivity axis	Range	Reported model response
Utilization u	0.90 → 0.70	Critical expansion speed falls from about 7% to about 3%.
Residual value s~	0.40 → 0.05	Maturity coverage falls from about 0.63 to about 0.08.
Debt cost r_d	7% → 15%	DSCR falls from about 1.00 to about 0.65.
Construction period τ	3 → 6 months	First-year FCF leakage roughly doubles.

The source describes the critical surface as nonlinear: simultaneous adverse movements in several variables can deteriorate FCF more quickly than a sum of single-variable linear extrapolations. Exact cell-by-cell matrices are not included in the supplied manuscript.

V. Calibration, Validation, and Model Limitations

5.1 Calibration comparison stated in the source

Model output	Public-data comparison cited in manuscript	Source interpretation
Baseline DSCR ≈ 0.8-1.0	CoreWeave first-year DSCR ≈ 1.2 with guarantees and SPV	Baseline understates coverage because guarantees and SPV structures are omitted.
r_d = 8.5%	CoreWeave blended cost ≈ 8.3%; bond yields as high as 15%	Stress scenario should use 12%-15%.
μ^* = 7%-13%	Industry capex growth 50%-90%	Gap direction is consistent.
Quarterly P(DSCR<1) = 26%-32%	Qualitative descriptions of Neocloud cash-flow volatility	Order of magnitude considered reasonable; precise comparison is absent.
s_L = 0.05; coverage ≈ 0.08	Used H100 three-year residual value 20%-30%	Model is conservative, but direction of step risk is consistent.

> **Evidence qualification**
>
> These comparisons are transcribed from the supplied manuscript. It does not include source citations, calculation exhibits, or a data appendix for the company-level benchmarks. They should be treated as calibration claims pending independent verification, not as independently validated facts.

5.2 Model limitations

1. Revenue is exogenous. R1 and p_{tok} are not endogenous to industry expansion; collective expansion could suppress prices, creating a fallacy-of-composition feedback that is not captured.
2. Heterogeneous agents are not modeled. Financing-capacity differences between hyperscalers and Neoclouds are not explicitly represented; λ and g only partially proxy for these differences.
3. Annual aggregation simplifies quarterly and monthly volatility using a normal approximation and does not include multi-period recursion.
4. Strategic responses are omitted. Firms may slow expansion, sell assets, or renegotiate debt when DSCR deteriorates.
5. Residual value is represented by a two-state discrete distribution rather than continuous GPU-generation decay.
6. Taxes and off-balance-sheet structures are simplified. τ_{tax} and λ are treated as constants rather than state-dependent variables.

VI. Consolidated Conclusion

The baseline closed-form solution provides a starting point: with R1 = 1,000, a 50% EBITDA margin, and C0 = 3,950, the critical expansion speed is approximately 7%-13%. The manuscript compares this with industry expansion of 50%-90%, implying a structural gap between operating cash generation and growth capex under the model assumptions.

Three binding layers beyond the baseline identity

1. Debt-service coverage is difficult to satisfy. Under the tested debt structures, the DSCR threshold of 1.20 requires utilization above 100%, even after reducing debt or extending amortization in the illustrated cases. The source interprets SPVs, guarantees, and long leases as mechanisms that make the structure marginally feasible.
2. Collateral refinancing can fail before annual cash flow does. Under low residual value, coverage falls to approximately 0.08 without a guarantee and 0.18 with a guarantee, both below the 1.20 refinancing threshold.
3. Timing and revenue mix create liquidity risk hidden by annual averages. A four-month delay is reported to consume roughly half of annual EBITDA under the narrower installed-base leakage calculation; quarterly DSCR-below-one probability is reported at 26%-32% under the stated normal approximation.

The model's central thesis is that financing-driven expansion arises when EBITDA relative to capital expenditure is systematically below the rate of expansion demanded by competition. Sustainability depends jointly on utilization recovery, residual-value stability, credit-spread stability, depreciation assumptions, and the refinancing window. The source positions the model as a diagnostic framework for the variables governing debt-chain resilience, not as a precise standalone default-probability estimator.

Appendix A. Reproduction Sheet

A.1 Baseline parameter set

Parameter	Value	Parameter	Value
C0	3,950	s	0.15
Tdep	6 years	R1	1,000
α	0.12	E	80.6
γ	0.30 (approx.)	EBITDA_1	500
Maintenance capex	2% of C0	b (levered case)	2,500
r_d	8.5%	DSCR covenant	1.20
β	0.05	s_ref	0.30
g	0.25	θ	1.20
Tloan tested	6 and 8 years	τ tested	4 months

A.2 Core formula sequence

1. Compute annual depreciation D = C0(1-s)/Tdep; adopt EBITDA_1 = 500 after the additional cost adjustment; derive unit EBIT and facility FCF.
2. Compute company FCF = n·EBITDA_1-m·C0-r_d·b·n, then solve the zero-FCF threshold μ*=(EBITDA_1-r_d·b)/C0.
3. For utilization scenarios, compute EBITDA_1(u)=499.4u-50 and DSCR(u)=EBITDA_1(u)/(r_d·b+A).
4. At maturity, compute V_eff(s~)/B_remaining and test the result against θ.
5. For the stochastic extension, draw u, s~, p_tok, and r_d jointly and calculate the four event probabilities specified in Section IV.

Implementation checks: reconcile maintenance-capex scope and construction-leakage calculations; define electricity-load conversions and the debt amortization schedule; document joint shock dependence and Monte Carlo settings; attach dated primary sources for calibration.

End of document.